Real-time spatial audio filtering for augmented-hearing wearables: Single-snapshot adaptive processing with imaginary boosting

Ryan Schmelzle [1], Meghan Parker [2] , Andy Choe [2]

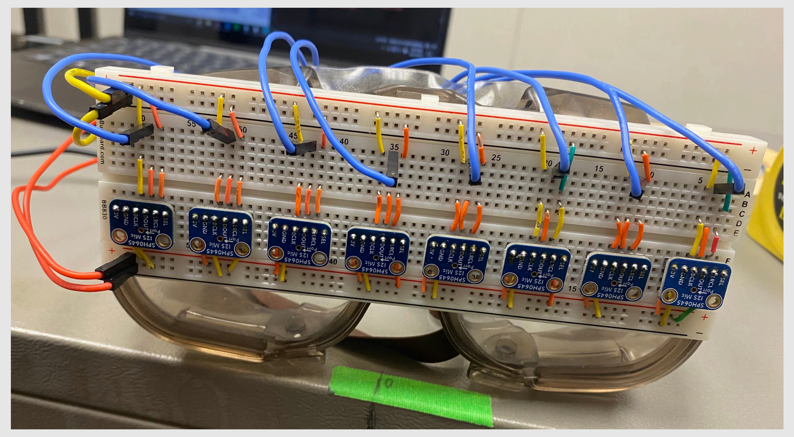
[1]Department of Computer Science, [2]Lampe Joint Department of Biomedical Engineering

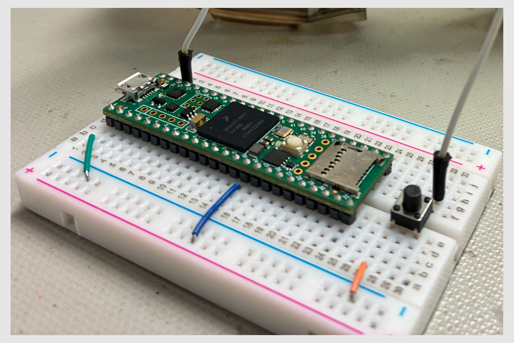
Phase 1: Platform Development

Background

- Noisy environments like restaurants and crowded events make it difficult to follow conversations; This is the “cocktail problem”
- Current hearing aids have small microphone arrays with limited spatial awareness; AR headsets prioritize vision over audio
- Glasses-frame wearables offer a wider microphone baseline, forward-facing coverage, and an everyday form factor
- This work presents a proof-of-concept beamforming system designed for the constraints of glasses-scale hardware, as a foundation toward a multimodal augmented-hearing platform
- **Our key contribution** is a new regularization technique called imaginary boosting that enhances the spatial resolution of microphone arrays beyond their physical size, enabling effective noise suppression on a glasses-scale device

System Overview

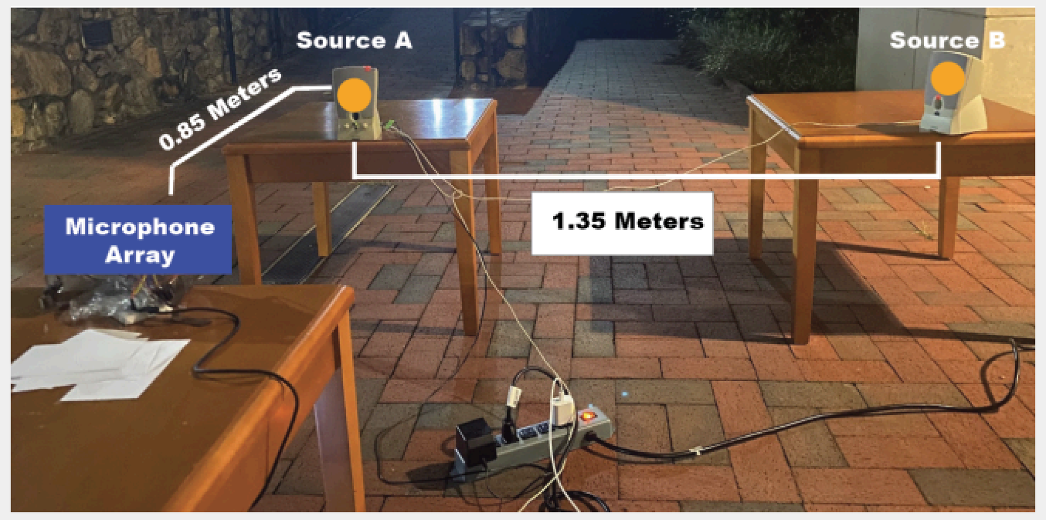




- 8 I2S MEMS Microphones
- 14.2 cm Array Span
- Teensy 4.1 Microcontroller
- 44.1 kHz Sampling Rate

Experimental Setup

- Designed a controlled testing scenario that emulates a typical outdoor restaurant seating scenario
- Two speakers to represent two adjacent patrons; one male, one female voice
- Microphone array represents an AR listener



Phase 2: Methods and Preliminary Results

Beamforming Algorithm

Isolate a sound source → Enhance target audio → Attenuate background noise

- Microphone array simultaneously samples audio across eight channels
- Constructs “steering vectors” to triangulate a sound source based on time delays between microphones
- Beamformer combines signal strength and directional information to isolate a target sound source

Novel Contribution: Imaginary Boosting

Problems with conventional beamforming

- The SCM cannot store enough information in any specific instant of time to accurately triangulate to a sound source
- Without regularization techniques, trying to isolate to a sound source is equivalent to trying to create a 3D sculpture from a singular 2D image

Imaginary Boosting

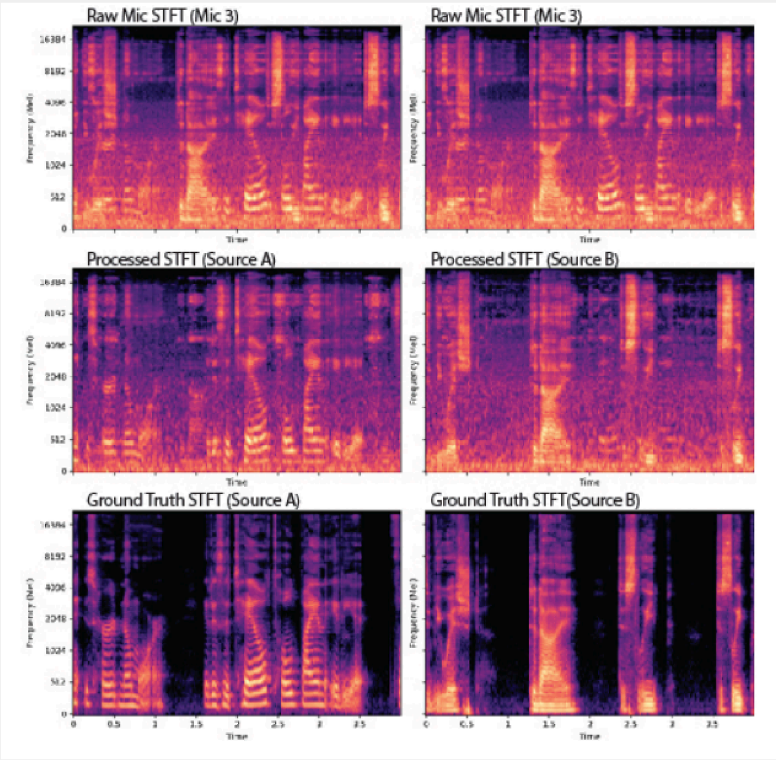
- Each number in the storage matrix contains real and imaginary parts

Real component: Sound signal strength

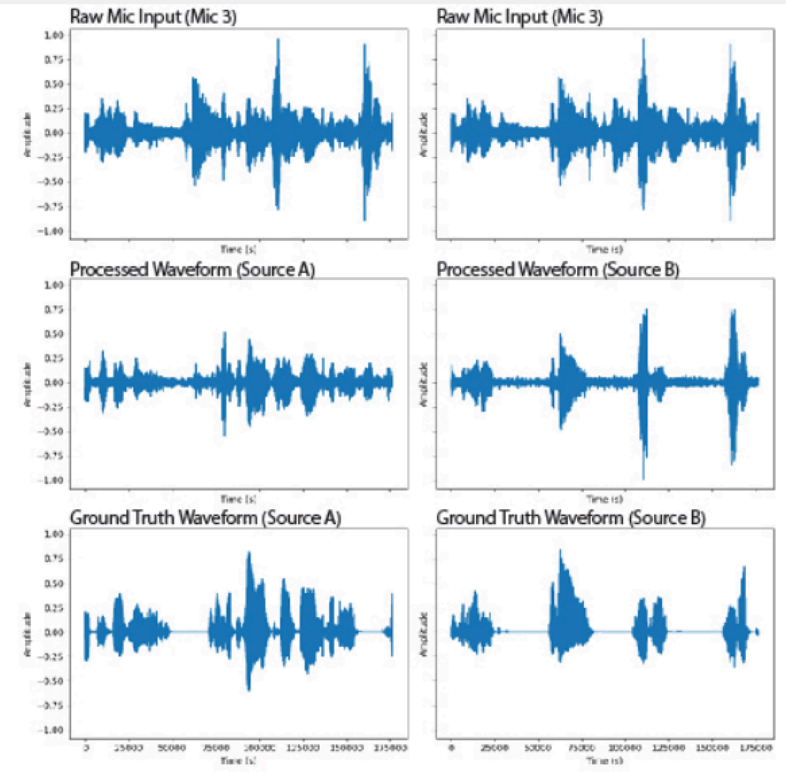
Imaginary component: Directional information ↑

- Our system boosts the imaginary component to artificially increase the size of the microphone array
- This gives us better spatial resolution and sound source isolation

Preliminary Results



Spectrogram: Processed audio time-frequency patterns converge toward ground truth, confirming effective source separation



Waveforms: Enhanced waveform peaks align more closely with ground truth than raw microphone input across both sources

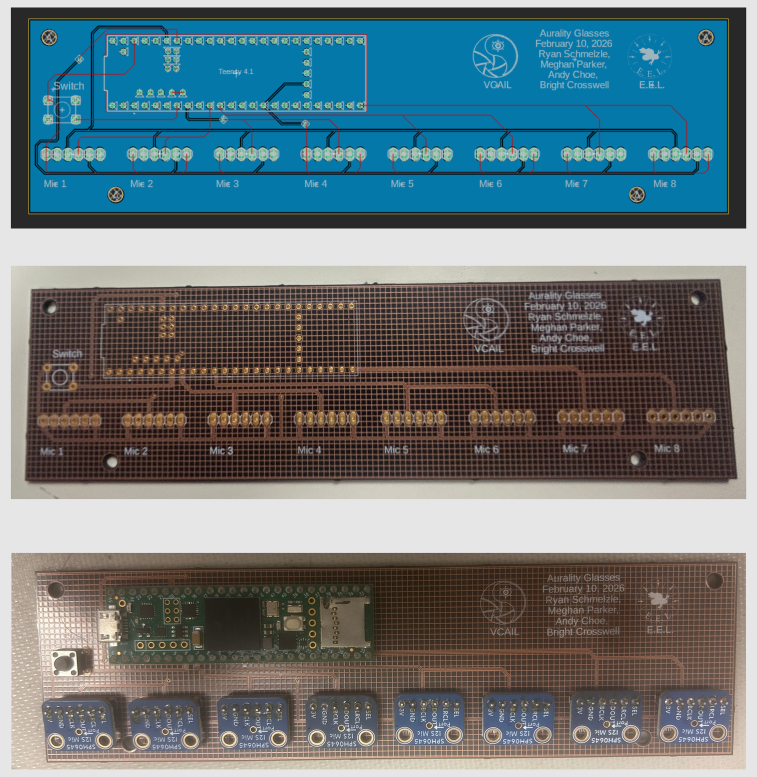
Phase 3: Wearable Integration

Custom Wearable Components

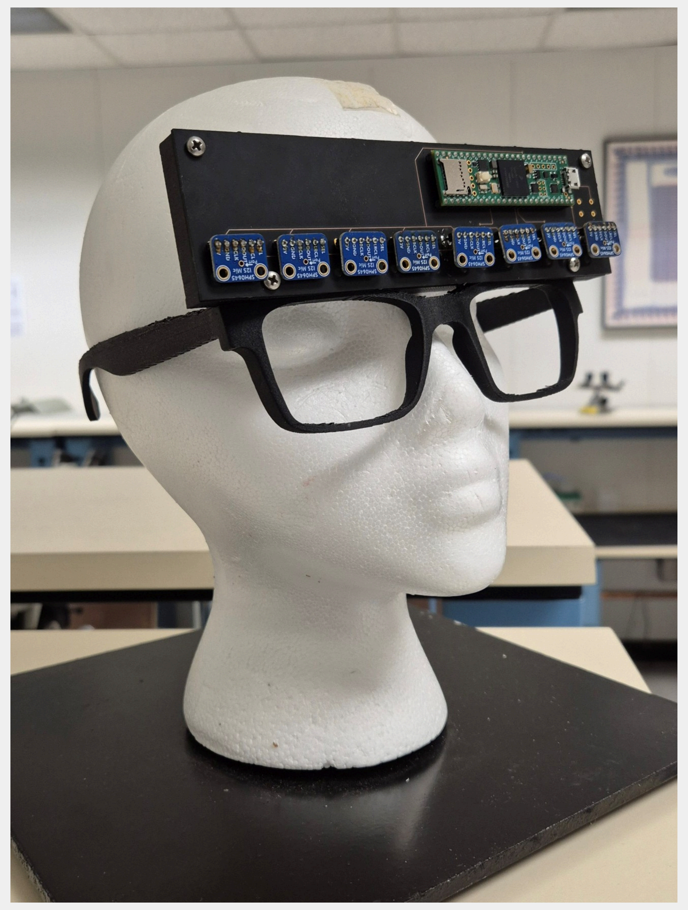
Glasses

The validated algorithm is deployed on the next prototype of the target wearable glasses form factor

Printed Circuit Boards



Testing Assembly



- An assembly with the custom glasses and PCB was constructed
- The assembly is intended for additional experimentation in the simulated restaurant scenario to replicate preliminary results with a wearable form factor
- Assembly implements the same linear array design as the proof-of-concept breadboard
- Space on the PCB is reserved for a mounted Intel RealSense D435i camera

Future Directions

- In order to emulate a truly wearable glasses form factor, the PCB footprint must be reduced
- Microphone surface mounts may be deconstructed and implemented directly into the board
- Smaller microcontrollers and cameras are an option for reducing assembly size

This work was conducted in the Experimental Engineering Lab (EEL) at UNC Chapel Hill. For additional information:
Principal Investigator Jim Mahaney: mahaney@cs.unc.edu
or
Meghan Parker: megapark@unc.edu